Mahdi Farmahini Farahani **(Arad)** **Ph.D. Student**
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EDUCATION

The Chinese University of Hong Kong (CUHK) Hong Kong SAR
Doctor of Philosophy , Geophysics (Seismology) *2026 – 2030*
Kharazmi University, Faculty of Earth Sciences Tehran, Iran
Bachelor of Science, Geology *2021 – 2025*

- Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025)
- Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)

University of Lille, International Academy Lille Hdf Lille, France
Pre-Ph.D. Water Resource Management Summer School (5 ECTS) *Summer 2024*

- Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia. Under Supervision of the Prof. Arnaud Gauthier.

TEACHING EXPERIENCE

CUHK Hong Kong SAR
Hydrogeology Teaching Assistant *August 2026 – Present*

- (EESC3320), under supervision of Prof. Yen Joe Tan

Space Research and Technology Institute (SRTI) Primorsko, Bulgaria
Lecturer *August 2026 – September 2026*

- EPEC ML WG Summer School 2026, under supervision of Prof. Lachezar Filchev

Iranian Young Scholars Club (IrYSC) Tehran, Iran
Earth Science Olympiad Teacher *April 2021 – August 2026*
Kharazmi University Tehran, Iran
Programming Teaching Assistant *April 2025 – June 2025*

- Programming Course, under supervision of Prof. Morteza Asemani

Kharazmi University Tehran, Iran
Geostatistics Teaching Assistant *February 2025 – June 2025*

- Geostatistics Course, under supervision of Prof. Ehsan Pegah

Young Scholars Club Tehran, Iran
Instructor *July 2024*

- Taught "Applied Data Science in Geology" for the Iran National IESO[1] 2024 Team

Kharazmi University Tehran, Iran
Instructor *April 2024 – June 2024*

- Organized and instructed the Advanced Python and Machine Learning Workshop

NODET[2] Tehran, Iran
Teacher *May 2021 – September 2021*

- Taught ITEO[3] Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools

RESEARCH EXPERIENCE

FEZtool Tehran, Iran
Co-Founder & Research and Development Specialist *August 2023 – Present*

- Led Earth science research using advanced algorithms and geospatial analysis.

Center for Complex Networks, SBU Partner Tehran, Iran
Research Assistant *November 2025 – June 2026*

- Worked on fashion trend forecasting project under supervision of Prof. Reza jafari.

International Astronomical Search Collaboration, NASA Partner Remote
Asteroid Search Team Member *July 2025 – September 2025*

- Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.

Kharazmi University Tehran, Iran
Research Assistant *October 2024 – January 2025*

- Performed MASW to extract dispersion images under supervision of Prof. Mehdi Talkhablou.

[1] International Earth Science Olympiad
[2] National Organization for Development of Exceptional Talents
[3] Iranian Thinking & Entrepreneurship Olympiad

COMMUNITY SERVICE

Zamin Pouya Scientific Journal Vol. 31 — Tehran, Iran
Associate Editor & Reviewer — *August 2025 – June 2026*

The 4th National Festival of Young Iranian Geologists — Tehran, Iran
Education Committee Member — *April 2025 – May 2025*

The 3rd National Festival of Young Iranian Geologists — Tehran, Iran
Graphics Committee Member — *May 2024 – June 2024*

Imam Khomeini International University — Qazvin, Iran
Editorial Board Member — *September 2023 – June 2024*

SELECTED PUBLICATIONS

Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. ***Kharazmi Journal of Earth Sciences.***
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. ***Ovan Scientific Journal***
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashefi, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. ***Journal of Water Management Modeling (JWMM)***

Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In ***Seismology Student Workshop (SSW 2025)***, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In ***The 1st Conference on Applied Geophysics in Mines***, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In ***2nd International Conference for Iranian Geography and Earth Sciences Students***, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In ***8th Symposium of Sedimentological Society of Iran***, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In ***26th Conference of Geological Society of Iran***, Urmia, Iran

Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence.* **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning.* **Tasir Publication.** (Original work published by Maurizio Petrelli)

PROJECTS

PlanetaryGeophysics | `github.com/aradfarahani/PlanetaryGeophysics` — *2025 – Present*

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

FEZrs (Open Source) | `pypi.org/project/fezrs` — *2023 – Present*

- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

Seismicity Analysis in Iran (2010–2025) | iris.edu/hq/workshops/2025/06 — *2025*

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

Geoelectricspy (Open Source) | `zenodo.org/records/10966815` — *2023*

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

INVITED TALKS

1. **Seismic Data Denoising in the Era of Neural Networks:** *5th National Festival of Young Iranian Geologists (Hosted by Mohaghegh Ardabili University, Kharazmi University, RIES, & UNESCO Chair);* 16 October 2026
2. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University;* 13 December 2025
3. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University;* 11 November 2025
4. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2;* 30 December 2024

LICENSES & CERTIFICATIONS

Applied Geotechnical Seismology, Kharazmi University *2026*

Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies, Kharazmi University

Seismology Skill Building Workshop 2025, EarthScope Consortium *2025*

Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method, National Iranian Geophysical Society

Remote Pilot Certification (Multirotor), Civil Aviation Authority of Islamic Republic of Iran *2023*

Unmanned Aerial Vehicle Surveying, Maan Academy

GIS, Mapping, and Spatial Analysis Specialization, University of Toronto (Offered by Coursera)

CS50 Certificate, CS50 (Harvard Uiversity) *2021*

HONORS & AWARDS

CUHK's International Student PhD Fellowship - *Awardee, CUHK* 2026

9th National Young Scientists Festival - *Merit Award Winner, Geology Category* 2025

Hydro 2024 Research-Oriented Summer School - *Full Scholarship*, University of Lille *2024*

Research Visit Grant (2500 DKK) - *Awardee*, University of Copenhagen *2022*

DrCT 2021 Global Computational Competition - *Gold Medalist* *2021*

1st national Thinking & Entrepreneurship Olympiad (ITEO) - *Silver Medalist* *2020*

WMI 2015 World Mathematical Championships Selection - *Silver Medalist* *2015*

SKILLS & INTERESTS

Programming Languages: Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R

Markup Languages: Markdown, LaTeX

Softwares: Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas

Field Techniques: Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition

Languages: Farsi (Persian): *Native*; English: *Fluent (C1)*; German: *Elementary (A2)*; Mandarin: *Beginner (Duolingo Score: 11)*

Research Interests: Signal/Image Processing, Geophysics (Seismology), Remote Sensing, Machine Learning, Open-Source Software Development

PROFESSIONAL MEMBERSHIPS

National Iranian Geophysical Society Student member, 2025 – Present

Space Generation Advisory Council Member, 2025 – Present

Google Maps Platform Innovators Member, 2025 – Present
Geological Society of Iran Student member, 2021 – 2022